Generative AI and Labor Displacement in the Global South:

Evidence from Philippine Business Process Outsourcing

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Abstract

Exported services have become for labor-abundant countries what manufacturing once was: a development path that absorbs large numbers of workers into internationally traded production. Generative AI performs many of the tasks these economies export, raising the possibility of labor displacement. I examine whether this displacement has begun in the Philippines, the economy most dependent on this development path, by combining pre-ChatGPT occupational composition with occupation-level observed AI usage from the Anthropic Economic Index and tracing labor market outcomes as U.S. client firms adopt AI. Employment growth slows in the most exposed provinces, industries, and municipalities, amounting to roughly a quarter million foregone jobs by mid-2025. The adjustment occurs through reduced hiring rather than layoffs: wages and hours do not change. Displaced workers enter unemployment, and the traditional adjustment through overseas migration does not offset these losses.

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1 Introduction

For labor-abundant countries, exporting services has become the development path that manufacturing once was. These countries sell customer service, records processing, and back-office operations to rich-country firms, and a recent line of work argues that these newly tradable services can carry the employment growth that factories no longer provide (Nayyar et al., 2021; Baldwin and Forslid, 2023; Rajan and Lamba, 2024; Rodrik and Sandhu, 2025). Generative AI (GenAI) threatens that path because the tasks these exports provide are the tasks the technology performs (Korinek and Stiglitz, 2021). The potential exposure is broad. The IMF estimates that GenAI affects about 40 percent of global employment, concentrated in the cognitive and clerical work that developing-country service exporters sell (Cazzaniga et al., 2024). The offsetting productivity gain may also be small. Acemoglu and Restrepo warn that AI directed at automating tasks rather than creating them delivers displacement without much productivity growth (Acemoglu and Restrepo, 2019, 2020), and macroeconomic accounting in the same framework puts the aggregate US productivity gains from GenAI below one percent over ten years (Acemoglu, 2025). Evidence on both productivity and displacement is only beginning to emerge. On the productivity side, the largest deployment study ran on offshore customer support agents and found gains of 15 percent, concentrated among novices (Brynjolfsson et al., 2025b). On the displacement side, the first estimates come from high-income economies: US payroll records (Brynjolfsson et al., 2025a), Danish administrative data (Humlum and Vestergaard, 2025), online labor platforms (Hui et al., 2024), and asset prices (Borri et al., 2026). All of this evidence comes from countries that buy the exposed work, and none from the countries that sell it, where the development

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strategy is at stake. Whether GenAI is displacing the labor that this development strategy relies on, at what margin, and with what consequences for displaced workers are the questions this paper takes up.

Offshore outsourcing provides a particularly clean setting because the shock arrives through foreign clients and exposure is predetermined. The Philippine Business Process Outsourcing (BPO) industry is the largest. The country has long exported workers, and its business process outsourcing (BPO) sector, employing 1.9 million workers for US clients (IT and Business Process Association of the Philippines, 2025), exports the work instead of the workers. India's technology industry is larger in absolute terms, at 5.4 million workers (NASSCOM, 2024), but those workers are below one percent of India's workforce (Government of India, 2024), against about four percent in the Philippines, so dependence is greatest there. It is also the largest source of formal private employment in the country, so what happens to it is a macroeconomic question rather than a sectoral one.

Detecting displacement this early requires knowing which places sell the exposed work and when their clients began to automate it. The measurement strategy has two parts. The first is a subnational measure of AI exposure. To construct it, I combine each province's occupational composition before the release of ChatGPT in November 2022, the start of mass-market GenAI, measured from labor force survey microdata by province of workplace, with occupation-level measures of *observed* AI usage from the Anthropic Economic Index (Handa et al., 2025; Massenkoff and McCrory, 2026). Whereas the existing exposure literature relies on predictions of what AI could do (Felten et al., 2021; Webb, 2020; Eloundou et al., 2024; Hampole et al., 2025), including a mapping of potential exposure for the Philippines itself (Cucio and Hennig, 2025), the weights I use measure what AI is observed to do.

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By this measure, customer service representatives, the core occupation of Philippine BPO, rank second in observed exposure among roughly 800 US occupations, with data entry keyers and medical records specialists close behind (Massenkoff and McCrory, 2026). The index locates the country's dependence on these occupations across the 87 province-level areas of the Labor Force Survey's workplace coding.

The second part is a measure of the treatment itself. Generative AI did not arrive on a single date: capability accumulated across successive model generations, and adoption diffused gradually into the firms that buy Philippine services. I therefore measure the shock as a dose, the share of US firms using AI in production, from the US Census Bureau's Business Trends and Outlook Survey (Bonney et al., 2024). This share rises from zero before late 2022 to more than seven percent of firms by early 2025. Because BPO demand is overwhelmingly American, US adoption is the relevant dose, external to Philippine provinces by construction. The estimating equation interacts predetermined provincial exposure with lagged US adoption while controlling for exposure-specific linear trends, so that identification comes from the curvature of the adoption path rather than from smooth differential trends in exposed places.

The mechanism that connects these two parts is straightforward. US firms from banks and insurers to hospitals and retailers buy customer service, data entry, and records processing from Philippine providers. When these firms adopt GenAI, they automate tasks that were previously purchased from Philippine providers, reducing demand for new seats. Providers respond as a high-turnover industry can, freezing hiring and letting attrition shrink headcount within a quarter or two. The result is displacement that arrives as foregone hiring in the provinces, industries, and municipalities where the exposed occupations concentrate.

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The displaced can then shift toward less exposed work within the sector, look elsewhere in the domestic labor market, or go abroad, and the paper follows each margin. The index measures where the pressure lands, and the adoption series measures when and how strongly it arrives.

The evidence indicates that labor displacement has begun. Employment declines with the adoption dose in every empirical design that I examine. As US adoption rises by one point, a province that is one standard deviation more exposed loses 0.15 percentage points of BPO employment share and 3.7 percent of BPO employment relative to less exposed ones. The same gradient appears within provinces across industries, in a specification where province-by-round fixed effects absorb every province-level shock, and it appears again across municipalities according to the intensity of their IT-zone infrastructure.

The magnitudes are economically meaningful. Between 2021 and mid-2025, BPO employment in the most exposed tercile of provinces grew by 42 percent, while the least exposed tercile grew by 67 percent. Had the most exposed tercile matched that pace, it would hold a quarter of a million more workers today, and cumulating the dose coefficient over the observed adoption path accounts for the whole of this gap.

The adjustment occurs through reduced hiring rather than layoffs. The response peaks at lags of one to two quarters and decays after that, as one would expect in an industry where attrition runs 30 to 40 percent a year, so a hiring freeze shrinks headcount within months without any contract being broken.[1] Consistent with this mechanism, wages and hours do not respond at all. The industry adjusts by reducing hiring rather than cutting

[1] Annual attrition in the Philippine BPO industry is commonly estimated at roughly 30–40 percent (Contact Center Association of the Philippines, 2024; Errighi et al., 2016).

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pay, the same entry-margin pattern found in US payroll data (Brynjolfsson et al., 2025a).

The displaced remain in the domestic labor market. Within the BPO sector, the occupation mix rotates away from the most exposed work, consistent with incumbents reassigned while hiring closes on entrants. Outside it, the would-be workers are so far absorbed by unemployment rather than by other employers: employment in other sectors of exposed provinces shows no response at lags of up to a year, the regional unemployment rate rises with the dose, and participation does not fall.

The overseas adjustment margin does not offset the domestic slowdown. In administrative deployment records covering 85 provinces of origin, first-time overseas departures fall with adoption in exposed provinces, and the decline is uniform across age groups, sexes, and destination countries. That pattern is not what displaced workers rerouting abroad would produce. It is most consistent with a shock to the household resources that finance migration. The margin that absorbed every previous shock to Philippine labor demand is, at this first reading, narrowing rather than opening.

The remainder of the paper examines whether GenAI has begun to weaken the service-led development path in the Philippines. Section 2 constructs the exposure index, Section 3 presents the empirical framework, Section 4 traces the labor market adjustment to AI adoption, and Section 5 concludes.

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2 An Observed-Usage Exposure Index for the Philippines

The index is a standard shift-share exposure measure. For province p,

$$\text{Exposure}_p \;=\; \sum_o s_{op}\, e_o, \tag{1}$$

where s_{op} is occupation o's share of employment among jobs located in province p, measured over the seven quarterly Labor Force Survey (LFS) rounds before the release of ChatGPT (2021Q2–2022Q4), and e_o is occupation o's observed AI exposure. The shares come from the LFS public use files (Philippine Statistics Authority, 2025). The LFS public use files record not only each employed person's occupation (2-digit PSOC) but also the province and municipality of the person's workplace. The index therefore measures the exposure of jobs located in a province rather than workers residing there. The weights e_o come from the Anthropic Economic Index's labor market release (Massenkoff and McCrory, 2026), which scores occupations by how much of their work AI is observed performing in millions of usage records. I crosswalk these scores from US occupation codes to the Philippine classification through ISCO-08. Supplemental Appendix Section S1 documents the crosswalk, whose coverage exceeds 99 percent of employment, together with an industry-level version of the index built from each industry's national occupation mix that carries the within-province analysis below, and Section S2 documents the harmonization of the index's successive releases. By that industry measure, business-support services, the division that contains call centers, is the most exposed of all 86 Philippine industries.

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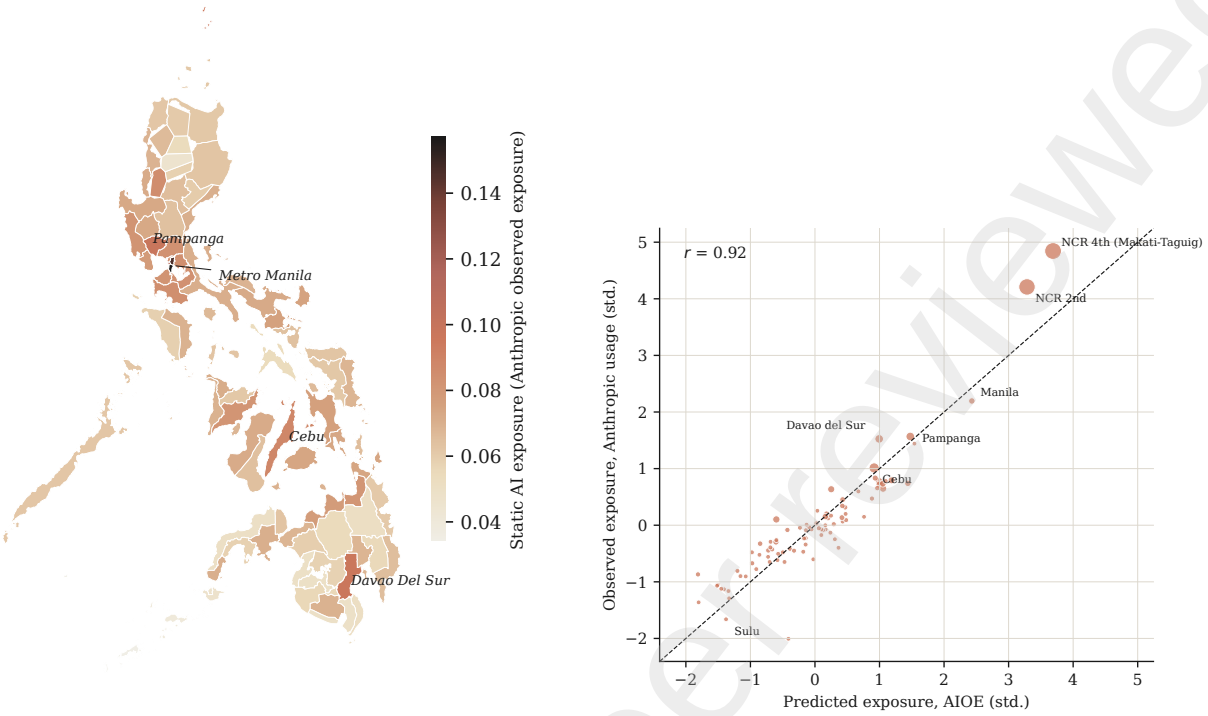
Using these weights for Philippine provinces requires one assumption. The relative ordering of occupations by AI substitutability must transfer from the usage data, which are global, concentrated in high-income countries, and organized on the US task taxonomy, to Philippine workplaces. In this setting the assumption is mild, because the occupations at the top of the ranking belong to an export industry whose jobs are American processes performed offshore: a customer-service job in Manila serves the same clients, follows the same scripts, and runs the same software as its US counterpart, often within the same firm. The assumption is weakest for non-traded occupations, where Philippine task content may depart from the US template. Measurement error of that kind attenuates the exposure gradient and works against the findings below. Panel B of Figure 1 provides the corresponding empirical check, since reweighting the index with an entirely different, capability-based scoring of the same occupations leaves the provincial ranking almost unchanged.

Panel A of Figure 1 maps the index. Exposure is highest where the BPO industry is concentrated: Metro Manila, led by the Makati–Taguig corridor, the Clark corridor in Pampanga, and the regional hubs of Cebu, Davao, Iloilo, and Bacolod. The agricultural and maritime periphery is lightly exposed. Panel B addresses the concern that this geography is an artifact of the usage-based weights. When I replace e_o with the capability-based AIOE of Felten et al. (2021) and rebuild the index (Supplemental Appendix Section S8), the resulting ranking is nearly identical ($r = 0.92$). The geography of exposure is therefore robust to how the occupation weights are constructed. The informative deviation is at the top of the distribution, where the usage-based index ranks the major hubs above the 45-degree line, so realized usage is more concentrated in their occupations than capability alone would predict.

Two additional patterns support the interpretation of the index. The first is that the

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(a) Provincial exposure

(b) Observed vs. predicted exposure

Figure 1: The provincial AI exposure index. Panel A: the static index, combining pre-ChatGPT (2021–2022) occupational employment by province of workplace (Labor Force Survey) with occupation-level observed AI exposure (Anthropic Economic Index). Panel B: the same index against a capability-based alternative that replaces usage weights with the AIOE of Felten et al. (2021). Both axes are standardized. Marker area is proportional to pre-period BPO employment. The dashed line is the 45-degree line.

industry-level version, which uses no geographic information, ranks office administrative and business support services, the division that contains call centers, as the most exposed of all 86 Philippine industries, and places gambling and betting second because the Philippine online-gaming industry staffs the same customer-service and back-office occupations (the full ranking is in Supplemental Appendix Table S8). The second is that the highly exposed provinces outside Metro Manila are the documented regional hubs of the industry, including the cities that the government's own digital-cities program designated as the sector's next

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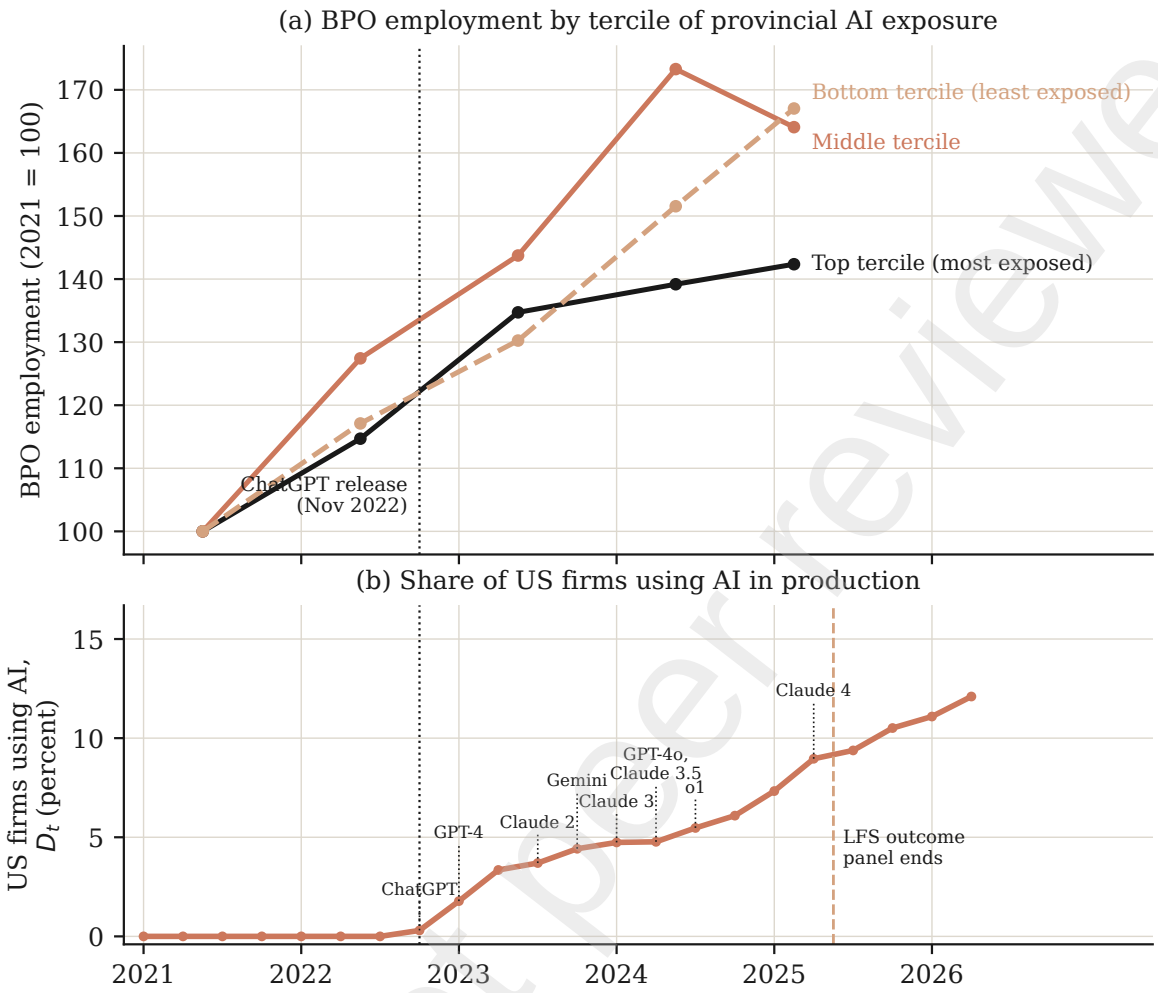


Figure 2: BPO employment by exposure tercile and the US adoption dose, on a shared timeline. Panel A: BPO covers PSIC divisions 62, 63, and 82, by province of workplace. Annual values average the quarterly LFS rounds (two rounds for 2025) and sit at year midpoints. Terciles split pre-period employment into thirds by the static index. Each series is indexed to its 2021 average. Panel B: D_t is the share of US firms using AI in production, from the Business Trends and Outlook Survey, with the early quarters spliced as described in Supplemental Appendix Table S3. Dotted markers indicate the release dates of major model generations. The dashed line marks the end of the LFS outcome panel in 2025Q2.

locations, rather than an arbitrary collection of urban areas.

Panel A of Figure 2 shows the first descriptive result that the index delivers. I divide provinces into employment terciles by exposure and track BPO employment in each group. The three groups grow at nearly identical rates through 2022, during the sector's post-

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pandemic boom, and diverge thereafter. By the first half of 2025 the least and middle exposed terciles stand at 167 and 164 percent of their 2021 levels, while the most exposed tercile stands at 142. The divergence begins between 2022 and 2023 and widens each year after that.

Panel B plots AI adoption by US client firms on the same timeline. The paper treats the shock as a dose rather than an event because capability accumulated across successive model generations, from ChatGPT in late 2022 through GPT-4, Claude 2, Gemini, Claude 3, and their successors, and no single release constitutes the treatment. Adoption instead diffused gradually into client firms, and the empirical analysis asks whether employment tracks this cumulative adoption.

3 Empirical Framework

The estimating equation is

$$y_{pt} \;=\; \alpha_p + \gamma_t + \beta \left(\text{Exposure}_p \times D_{t-k}\right) + \delta \left(\text{Exposure}_p \times t\right) + \varepsilon_{pt}, \tag{2}$$

where y_{pt} is a BPO employment outcome for province p in quarterly LFS round t (2021Q2–2025Q2), Exposure_p is the index of equation (1), standardized to mean zero and unit variance across provinces weighted by pre-period employment, and D_t is the share of US firms using AI in production. The series comes from the biweekly Business Trends and Outlook Survey aggregated to quarters (Bonney et al., 2024). I splice the first three post-release quarters with payment-platform and web-traffic data and set the series to zero before 2022Q4 (construction

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in Supplemental Appendix Table S3). Alternative splices and client-weighted doses, by sector and geography, leave the estimates unchanged (Tables S1 and S2). Worker-side surveys record the same rapid diffusion (Bick et al., 2024).

US adoption, rather than Philippine usage, measures the shock because BPO demand is American and arrives through clients' decisions. The Anthropic data identify the cross-sectional exposure, while the adoption survey provides the time-series variation. The exposure-specific linear trend ensures that β is identified by the curvature of the adoption path, so that any confound trending smoothly in more exposed places, such as urbanization, sectoral drift, or industry maturation, loads onto δ rather than onto β. Adoption enters at a lag k because it reaches Philippine payrolls through hiring freezes, attrition, and contract renewals. The baseline uses $k = 1$, and the results trace the full lag profile.

The identifying assumption at the province level is that, conditional on exposure-specific linear trends, no omitted shock tracks the shape of the adoption curve differentially across exposure. Two additional designs relax this assumption. The first estimates equation (2) across industries within provinces, replacing Exposure_p with the industry-level index and including province-by-round and province-by-industry fixed effects. This specification absorbs every province-level shock, including local demand and weather, so identification comes from differences in exposure across industries within the same province and quarter. Shocks specific to exposed industries, such as the 2023 return-to-office mandate for tax-zone firms, are not absorbed by construction. Supplemental Appendix Section S4 shows that the mandate's step-like timing does not track the adoption curve, its late-2024 relaxation coincides with growing rather than shrinking effects, and it provides no explanation for the within-sector occupation rotation reported below.

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The second design estimates equation (2) across municipalities, interacting the number of a municipality's IT-zone Presidential Proclamations through 2021 with the adoption dose and including municipality and province-by-round fixed effects. Supplemental Appendix Table S4 tests the identifying assumption directly in the pre-release rounds using a joint curvature test and a placebo dose placed where none existed. The within-province and municipal designs pass both tests, while the province-level gradient does not because the pre-release rounds span the reopening from pandemic restrictions that bound hardest in the capital. The province-level estimates therefore carry the descriptive weight, while the within-province and municipal designs carry the identification.

Standard errors are clustered at the treated unit throughout. The regressions are weighted by pre-period employment, fixed before the release, so that the estimates average over workers rather than areas and downweight the smallest, noisiest survey cells. The industry specification also reports unweighted estimates, since a gap between the two would indicate heterogeneous treatment effects (Solon et al., 2015); the two are nearly identical. Shock-level inference in the sense of Adão et al. (2019), implemented through the equivalence of Borusyak et al. (2022), yields smaller standard errors than those reported here (Supplemental Appendix Table S6).

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4 Results

4.1 Employment

Table 1 reports the paper's central estimates. Columns 1 and 2 show the province-level gradient. A province that is one standard deviation more exposed loses 0.149 percentage points of BPO employment share (SE 0.027) and 3.7 percent of BPO employment (SE 1.2) for each point of lagged US adoption, conditional on exposure-specific trends. These estimates describe where the slowdown is concentrated, but the pre-release tests in Supplemental Appendix Table S4 indicate that they partly reflect the capital's reopening path, so the identification rests primarily on the designs that follow. Within provinces, an industry that is one standard deviation more exposed loses 0.6 percent of employment per adoption point in the weighted specification and 1.2 percent in the unweighted specification (columns 3 and 4). Because these estimates compare industries within the same province and quarter, province-by-round fixed effects absorb the reopening cycle together with every other province-level shock. Municipalities show the same pattern at the finest level of geography: employment falls in municipalities with greater IT-zone infrastructure relative to other municipalities in the same province as adoption accumulates (column 5; variants in Supplemental Appendix Table S7). Across three designs with different sources of identifying variation, the estimated effects are consistently negative and statistically distinguishable from zero.

The magnitudes are economically meaningful, and Panel A of Figure 2 provides a natural benchmark. A province at the 75th percentile of exposure and one at the 25th percentile differ by 0.71 standard deviations. Cumulating the column 2 coefficient over the observed adoption path implies that the more exposed province holds 19 log points less BPO employ-

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Table 1: Employment responses to AI adoption across designs

	(1) BPO share	(2) log BPO	(3) log emp	(4) log emp	(5) log BPO	(6) log pay	(7) log hours
Exposure × adoption (lag 1)	-0.1495	-0.0366				0.0015	0.0021
	(0.0275)	(0.0119)				(0.0040)	(0.0019)
Exposure × adoption (lag 1)			-0.0060	-0.0123			
			(0.0025)	(0.0027)			
Exposure × adoption (lag 1)					-0.0595		
					(0.0258)		
Observations	1,479	1,312	67,290	67,832	1,170	594	594
R^2	0.977	0.953	0.960	0.902	0.894	0.655	0.378

Notes: The reported regressor is exposure interacted with US AI adoption lagged one quarter. All columns include exposure-by-linear-trend interactions and unit and time fixed effects. Columns 1–2: province panel (87 provinces, 17 rounds), weighted by pre-period employment, standard errors clustered by province. Columns 3–4: province-by-industry panel (86 industries) with province-by-round and province-by-industry fixed effects. Column 3 is weighted by pre-period cell employment. Standard errors are clustered by province-industry. Column 5: municipality panel with IT-zone intensity (log of one plus pre-2022 IT-zone proclamations) as exposure, municipality and province-by-round fixed effects, standard errors clustered by municipality, unweighted (weighted and clustering variants in Supplemental Appendix Table S7). Columns 6–7: province-level BPO daily pay and weekly hours (cells with at least ten sampled BPO workers). Standard errors in parentheses.

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ment by mid-2025 through the adoption channel alone. Scaled to the tercile comparison, the province-level estimate accounts for the entire 16-log-point growth gap between the most and least exposed terciles, though it modestly overshoots the observed gap. That overshoot likely reflects the pre-release reopening curvature documented in Supplemental Appendix Table S4, which biases the province-level estimates downward. The within-province specification, which passes the pre-release tests, yields nearly identical magnitudes. The BPO divisions sit 3.6 standard deviations above the employment-weighted mean industry, so the estimated coefficient cumulates to a 16-log-point shortfall by mid-2025, matching the tercile comparison. In levels, had the most exposed tercile grown at the rate of the least exposed tercile from 2021 onward, it would employ roughly 240,000 additional BPO workers today—about one-eighth of the sector. The adjustment therefore takes the form of foregone employment growth rather than layoffs.

4.2 Timing

Figure 3 examines how quickly employment responds to client adoption. If the estimates reflect the transmission of client adoption to Philippine hiring, the response should arrive with a lag of a quarter or two rather than contemporaneously, because the industry's structure makes slowed hiring the least costly adjustment margin on both sides of the contract. Panels (b) and (c) confirm this prediction. The response peaks after one to two quarters in every design and attenuates by the fourth quarter. The province-level coefficient is -0.115 contemporaneously and -0.150 at a one-quarter lag, while the weighted within-province estimate turns on at lags one and two. The lag profile is also difficult to reconcile with confounding,

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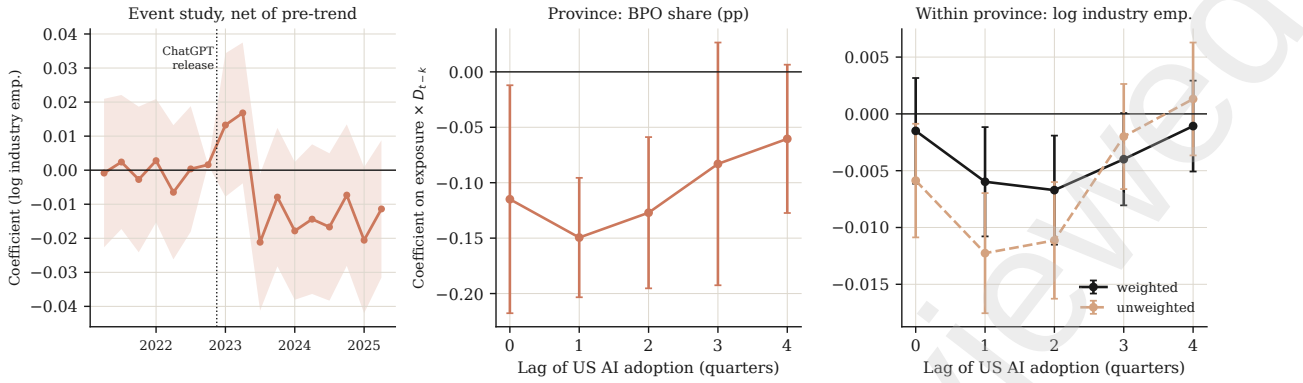


Figure 3: Dynamics of the employment response. Panel (a): within-province industry event-study coefficients, net of a linear trend fitted to the six pre-release points (specification of Table 1, column 3). Panels (b) and (c): each point is a separate regression with US adoption at the indicated lag, specifications of Table 1, columns 1 and 3–4. Bars: 95 percent confidence intervals. Panel (a) deviations are individually imprecise, and the powered test is the dose specification of Table 1.

because a smooth omitted shock trending with adoption would not generate a peaked and decaying response.

Panel (a) presents the same pattern in event-study form after removing the fitted pre-release trend. The coefficients remain close to zero before the release of ChatGPT and during the first two post-release quarters, when lagged adoption was still negligible, before turning negative from mid-2023 onward. Individual quarter estimates are imprecise because the trend is estimated from only six pre-release observations, so the powered test remains the dose specification of Table 1. Throughout this period, employment in the exposed industries continues to grow in levels. The response is therefore a slowdown in hiring rather than an outright contraction.

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4.3 The Hiring Margin

Columns 6 and 7 of Table 1 apply the same specification to BPO daily pay and weekly hours, and neither outcome responds. The pay coefficient of 0.001 (SE 0.004) rules out wage responses larger than one percent per adoption point. Foregone entry also removes junior workers from the mix, raising observed pay, so this bound is conservative. The sector is adjusting through the quantity of new hiring rather than the price of incumbent labor. This pattern is consistent with the lag structure in Figure 3 and mirrors the early evidence from US payroll records, where the initial effects of GenAI fall on would-be entrants rather than incumbent workers (Brynjolfsson et al., 2025a). For a workforce as young as the Philippine BPO sector, foregone entry is where labor displacement first appears.

4.4 Where the Displaced Workers Go

The employment estimates raise the question of where displaced workers adjust. The first margin lies within the sector itself. Within the BPO divisions, the occupation mix rotates away from high-exposure occupations as adoption rises, by about 7 percent per exposure standard deviation per adoption point (Supplemental Appendix Table S13), consistent with incumbents being reassigned while hiring closes on entrants. The aggregate employment estimates therefore net these offsetting adjustments within the sector.

For workers the sector no longer absorbs, the remaining margins are employment in other industries, unemployment, labor force participation, and overseas work. The LFS records industry at the province of workplace, while unemployment and participation are recorded at the region of residence. I therefore estimate sectoral adjustment across the 87 province-level

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areas and unemployment and participation across the 17 regions using employment-weighted regional exposure. National labor market conditions provide useful background. Between early 2021 and mid-2025, the unemployment rate fell from 8.8 to 4.1 percent while labor force participation remained between 60 and 65 percent, so any increase in unemployment among exposed regions occurs against a strengthening aggregate labor market.

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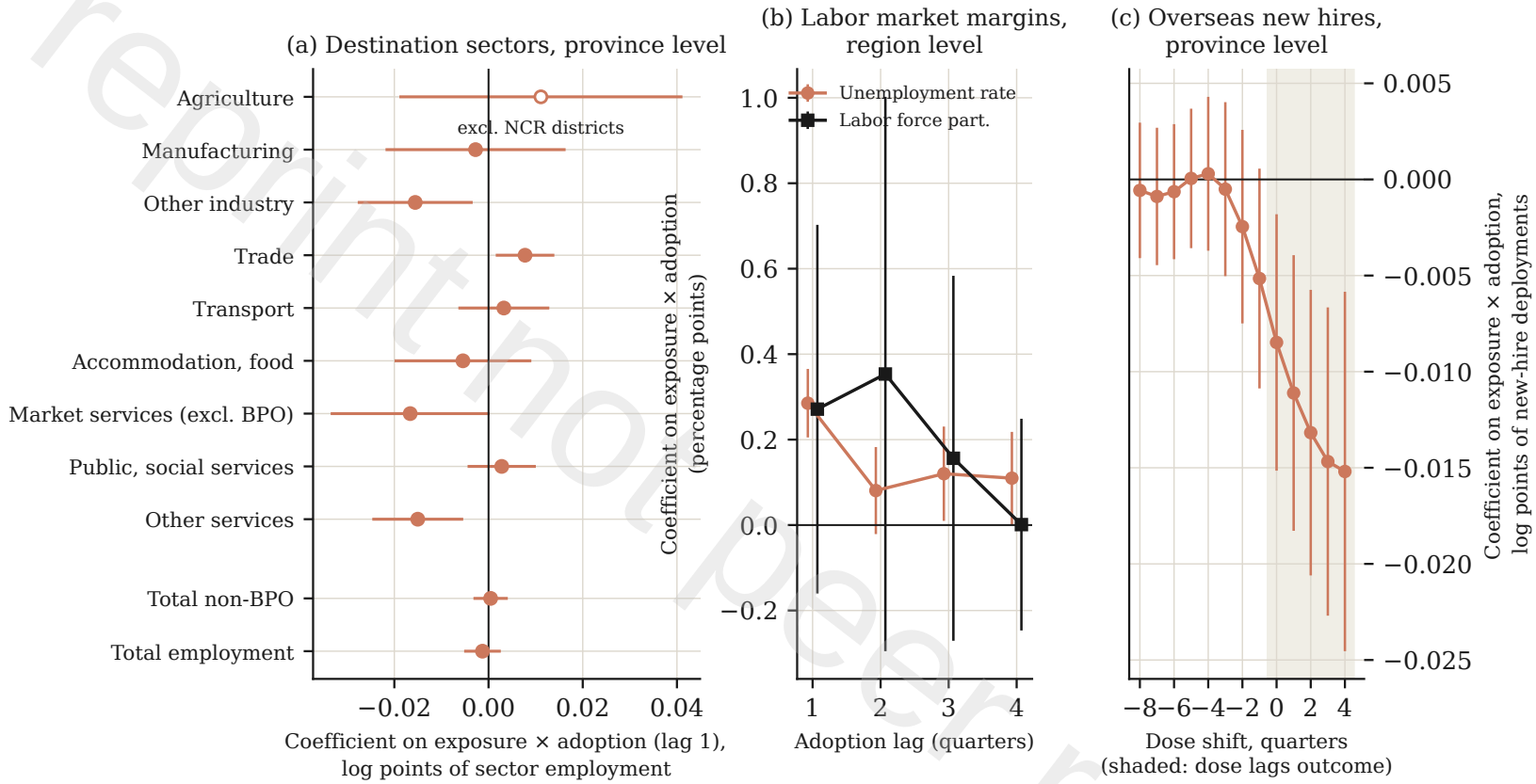


Figure 4: Where the adjustment goes. Panel (a): coefficients on exposure interacted with US adoption at a one-quarter lag, from equation (2) estimated on log employment in each destination sector across all 87 province-level areas, with BPO divisions 62, 63, and 82 excluded throughout. The agriculture estimate (open marker) instead excludes the four NCR districts, whose small agricultural cells otherwise dominate it. Both samples for every sector are in Supplemental Appendix Table S9. Panel (b): the same specification across the 17 regions, replacing provincial exposure with its employment-weighted regional average, for the unemployment rate and the labor force participation rate in percentage points, with adoption entering at lags of one to four quarters. Panel (c): coefficients on exposure interacted with US adoption on log first-time overseas deployments, from the province-by-quarter panel of Department of Migrant Workers records, with the adoption series shifted from eight quarters ahead to four quarters behind on a fixed estimation sample and no trend control (Supplemental Appendix Table S10). Bars: 95 percent confidence intervals. Standard errors clustered by province in Panels (a) and (c) and by region in Panel (b).

Panel (a) of Figure 4 applies equation (2) to employment in nine destination sectors, excluding the BPO divisions. No sector absorbs the displaced workers. Every coefficient lies within 0.02 log points of zero. The only positive estimate distinguishable from zero, in trade, is small (0.008, SE 0.003), disappears by a one-year lag, and does not survive excluding the NCR districts. The remaining significant estimates are small negatives in other industry and other services, consistent with mild local spillovers rather than reallocation.

Total employment summarizes the sectoral margin. Because BPO accounts for about 4 percent of employment in the panel, complete displacement without local reallocation would imply a decline of about -0.0015 log points in total employment per adoption point. The estimated effect is -0.0013 (SE 0.0020), remarkably close to this benchmark. Although the estimate cannot statistically distinguish complete from partial reallocation, its magnitude is consistent with little short-run absorption into other sectors. Reemployment takes time, yet the sector coefficients remain close to zero even at lags of up to one year.

Panel (b) traces unemployment and labor force participation across adoption lags. Unemployment rises by 0.29 percentage points per adoption point per unit of regional exposure at a one-quarter lag (SE 0.04, wild cluster bootstrap $p = 0.013$), and the point estimates remain positive at every horizon through one year, though the longer lags become imprecise under the bootstrap. Participation does not respond at any horizon. Taken together, these results indicate that displaced workers remain in the labor force and queue as unemployed rather than finding employment elsewhere. The unemployment magnitude is also consistent with the foregone-jobs arithmetic of the previous section. Cumulated over the observed adoption path, the coefficient implies that the National Capital Region, which stands about two units of regional exposure above the other regions, carries an unemployment rate sev-

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eral percentage points higher by mid-2025 than the exposure of other regions would imply. An alternative calculation that allocates roughly half of the quarter million foregone BPO jobs to the capital, relative to its labor force of about seven million, produces a similarly sized effect. Both calculations place the response in the low single digits, suggesting that the unemployment estimates are economically plausible. The full set of estimates appears in Supplemental Appendix Table S9.

4.5 The Overseas Margin

Overseas migration has long been the Philippines' principal adjustment margin to weak domestic labor demand, with remittances insuring households against local income shocks (Yang and Choi, 2007). The BPO industry emerged as an alternative by allowing workers to earn foreign wages without leaving the country (Baldwin and Forslid, 2023; Blinder and Krueger, 2013). If GenAI reduces demand for these jobs, displaced workers may instead enter the overseas labor market. Whether they do so is an empirical question. To examine this margin, I use administrative records from the Department of Migrant Workers containing monthly deployment counts by province of origin, separately identifying first-time deployments and rehires, together with annual breakdowns by age, sex, and destination.

First-time overseas departures fall. In the province-by-quarter panel of deployment counts (Department of Migrant Workers, 2026), first-time deployments decline by 2.0 percent per adoption point per exposure standard deviation (wild cluster bootstrap $p = 0.004$ on 85 province clusters), while rehires, workers already in the overseas labor market, do not respond, so the share of first-time migrants in total deployments also falls (Supplemental

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Appendix Table S10). The timing matches the proposed mechanism. Shifting the adoption series relative to the deployment panel yields no effect four to eight quarters before the shock, while the decline begins when adoption arrives and grows over the following year (Panel (c) of Figure 4). The stock of overseas workers shows no response over the same horizon (Supplemental Appendix Table S9), as one would expect for a slow-moving stock (Theoharides, 2018). Because deployment is recorded at the province of origin whereas exposure is measured at the province of workplace, this geographic mismatch attenuates the estimates, so the true effect is likely larger.

The annual breakdowns show that the decline is broad-based. It is no larger among younger workers than older ones, no larger among women despite the industry's female skew, and no more pronounced across destination corridors (Supplemental Appendix Tables S11 and S12). This broad pattern is consistent with a shock to the household resources that finance migration rather than one concentrated in particular demographic groups or destination markets. Migration is a costly investment, and negative household income shocks reduce departures just as positive transfers increase them (Bazzi, 2017; Angelucci, 2015). This interpretation is also consistent with experimental evidence from the Philippines showing that neither lowering information costs nor improving access to overseas recruiters substantially increases migration (Beam et al., 2016; Beam, 2016), suggesting that overseas employment is constrained primarily by household resources.

One qualification is that the estimation window coincides with the recovery from the pandemic collapse in overseas deployments, and the most exposed provinces recovered earliest. Supplemental Appendix Table S5 reports four checks of this concern. The decline survives aggregation to regions, neighboring provinces do not gain what exposed provinces lose, and

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excluding 2022, when the differential rebound is sharpest, leaves the estimate largely unchanged. A placebo dose assigned to 2018–2019 returns an imprecise coefficient of similar magnitude, suggesting some caution in interpreting the size of the decline. The broader policy implications of these findings are discussed in Supplemental Appendix Section S9.

5 Conclusion

Exported services have become for labor-abundant countries what manufacturing once was: a way to sell labor to the world without moving workers across borders. Generative AI performs many of the tasks that underpin this development path. This paper provides early evidence that the resulting labor displacement has begun in the Philippines, the economy most dependent on exporting those tasks. Although the BPO sector continues to grow, employment growth slows systematically where exposure is greatest. The adjustment occurs through reduced hiring rather than layoffs, tracks the accumulation of AI adoption among US client firms with the short lag implied by hiring freezes and attrition, and leaves wages and hours unchanged. Between 2021 and mid-2025, this slowdown amounts to roughly a quarter of a million foregone BPO jobs in the most exposed provinces.

The paper also traces where the displaced workers go. Within the sector, employment shifts toward less exposed occupations. Outside it, workers remain unemployed rather than moving into other industries or leaving the labor force. Nor does the traditional overseas adjustment margin expand: first-time overseas departures fall in the most exposed provinces, suggesting that the income shock associated with weaker BPO hiring also reduces the household resources needed to finance migration abroad. At least over the horizon studied here,

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the adjustment therefore occurs primarily within the domestic labor market.

US adoption continues to rise, so the adjustment documented here is unlikely to be complete. If the mechanism is reduced hiring rather than layoffs, the effects should spread gradually across cohorts as fewer entrants replace departing workers. The Philippine evidence therefore suggests that the first effects of GenAI on the service-led development path emerge through slower entry rather than mass displacement. Employment continues to grow, but it grows least where exposure is greatest; the workers the sector no longer absorbs queue as unemployed rather than moving into other sectors or overseas employment. The development strategy remains viable, but its capacity to absorb new workers has begun to narrow.

Data availability

The replication package deposits the code and data that reproduce every result. Measures are constructed from public sources, the Anthropic Economic Index (Anthropic, 2026), Philippine Labor Force Survey public use files (Philippine Statistics Authority, 2025), the registries of the Philippine Economic Zone Authority (Philippine Economic Zone Authority, 2021), Philippine Presidential Proclamations (Official Gazette of the Republic of the Philippines, 2021), and the US Census Bureau's Business Trends and Outlook Survey (US Census Bureau, 2025), together with provincial deployment records provided on request by the Department of Migrant Workers (Department of Migrant Workers, 2026). The survey public use files and the deployment workbook cannot be redistributed. The package includes the intermediate and analysis datasets built from them and the code that builds them, so all results replicate without the restricted files.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic), across the Claude Opus and Claude Sonnet model families, in order to write and debug the data assembly, analysis, and plotting code contained in the replication package, to assist in constructing the occupation and geographic crosswalks, and to improve the clarity and language of text drafted by the author. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Supplemental Appendix

Generative AI and Labor Displacement in the Global South:
Evidence from Philippine Business Process Outsourcing

Nico Ravanilla

August 4, 2026

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S1 Crosswalk Construction and Coverage

The occupation weights originate in the 2018 US Standard Occupational Classification (SOC), whereas the Labor Force Survey codes occupations using the Philippine Standard Occupational Classification (PSOC 2012). I connect the two through ISCO-08, mapping each 2018 SOC occupation to its 2010 SOC predecessors and then to ISCO-08 unit groups using the official BLS crosswalks. Because a single SOC occupation can map to several ISCO unit groups, I split its May 2023 OEWS employment (US Bureau of Labor Statistics, 2023) equally across the mapped groups and define each ISCO unit group's exposure as the employment-weighted mean exposure of its constituent occupations. The procedure covers 421 of the 436 ISCO-08 unit groups. Because the LFS public use files record PSOC at two digits, the analysis operates at the ISCO-08 sub-major level, where PSOC 2012 and ISCO-08 coincide. The 87 province-level areas of the LFS workplace coding are the 81 provinces of that classification, the four districts of Metro Manila, and the independent cities of Cotabato and Isabela. As a final coverage check, the share of provincial employment matched to an exposure weight averages 100 percent across the 87 areas, with a minimum of 99.8 percent.

S2 Anthropic Economic Index: Release Harmonization

The Anthropic Economic Index has been released in successive waves whose schemas and classifiers differ, and the replication package harmonizes six usage windows spanning December 2024 to May 2026, separately for the consumer product (Claude.ai) and the first-party API. Usage records are matched to O*NET tasks in each release, with essentially complete coverage. Within each release I renormalize usage shares to the classified total, since the share of unclassified usage varies across releases. Two caveats matter for anyone using the harmonized panel. First, there are level breaks between release families, which reflect classifier and taxonomy changes rather than changes in usage, so growth should be computed within a release family or absorbed with window effects. Second, the country-level data cover the consumer product only, while enterprise API usage, where automation-type use concentrates, is published at the global level. Neither caveat affects the main estimates in the paper, because the paper uses only the time-invariant observed-exposure weights from the labor market release, and the time variation comes from the adoption survey instead.

S3 The Adoption Dose: Construction and Alternatives

Table S3 reports the quarterly dose series and how each segment is constructed. The core of the series is the US Census Bureau's Business Trends and Outlook Survey, which asks firms biweekly whether they used AI in producing goods or services and which begins fielding that question in September 2023. Two adjustments extend the series. First, I set the series to zero before 2022Q4 and backcast the three quarters from 2022Q4 to 2023Q2 by scaling the first BTOS reading with the growth of the Ramp AI adoption index (Ramp, 2025), cross-checked against ChatGPT web-traffic estimates and Google Trends. Second, the BTOS question changed in November 2025 from

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AI use in production to AI use in any business function, producing a level jump from 9.9 to 17.3 percent. I place the revised readings on the original basis using the ratio of the trend-extrapolated original series to the first revised observation, which equals 0.600. Three alternative constructions leave the employment results in main-text Table 1 unchanged. The first sets the backcast segment to zero. The second uses the revised post-November-2025 readings without rescaling and absorbs the wording change with an indicator. The third ends the sample in 2025Q3, before the question change, which is also before the end of the LFS outcome panel used in the paper.

A fourth variant addresses the fact that BPO clients are not average US firms. The BTOS publishes adoption by sector, and the IT-BPM industry's client verticals concentrate in the sectors that adopted fastest. I therefore construct a client-weighted dose that averages sector-level BTOS adoption using the industry's global client-vertical mix from the IBPAP Roadmap 2028: banking, financial services, and insurance at 26 percent mapped to NAICS 52, technology, media, and telecom at 20 percent mapped to NAICS 51, retail at 10 percent mapped to NAICS 44, manufacturing at 11 percent mapped to NAICS 31, healthcare at 11 percent mapped to NAICS 62, energy at 6 percent split between NAICS 21 and 22, and the remaining 18 percent carrying the national series. The client-weighted dose runs about 40 percent above the national series at every point because the client sectors adopt faster, and it correlates with the national series at 0.997, so the shape that identification relies on is unchanged. Table S1 re-estimates the four employment columns of main-text Table 1 with this dose at a one-quarter lag. The coefficients scale down in proportion to the higher dose level, while the test statistics are unchanged or stronger.

A fifth variant addresses client geography rather than client sector. North America accounts for about 70 percent of the Philippine industry's buyer base, with Europe including the UK at about 15 percent and APAC at about 15 percent, from the same IBPAP Roadmap 2028. Workers serving non-US clients faced those countries' adoption paths rather than the US path, so the US-only dose overstates their treatment intensity and dilutes the estimated response toward zero. I construct a geography-weighted dose that averages three blocks by these shares. The North America block is the US series. The Europe block is the share of UK businesses using large language models for text generation from the ONS Business Insights and Conditions Survey, the only national series that measures generative AI use specifically. The APAC block is the mean post-2022 adoption increase in Japan's Communications Usage Trend Survey and Singapore's IMDA enterprise survey. Each foreign block is zero through 2022Q4, matching the splice convention of the main dose, and annual readings are interpolated to quarters. The resulting geography-weighted dose correlates 0.998 with the US-only series. Table S2 re-estimates the four employment columns with this dose at a one-quarter lag. The coefficients are modestly larger than under the US-only dose, consistent with the foreign blocks rising somewhat faster late in the sample, and the test statistics are unchanged.

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Table S1: Client-weighted dose: employment responses

	(1) BPO share	(2) log BPO	(3) log emp	(4) log emp, unwtd
Exposure × client-weighted adoption (lag 1)	-0.1050	-0.0255		
	(0.0190)	(0.0081)		
Exposure × client-weighted adoption (lag 1)			-0.0043	-0.0089
			(0.0017)	(0.0019)
Observations	1,479	1,312	67,290	67,832
R^2	0.977	0.953	0.960	0.902

Notes: Same specifications as main-text Table 1, columns 1 to 4, with the national adoption dose replaced by the client-weighted dose at a one-quarter lag. Columns 1 and 2 use the province panel with province and round fixed effects, weighted by pre-period employment, with standard errors clustered by province. Columns 3 and 4 use the province-by-industry panel with province-by-round and province-by-industry fixed effects, column 3 weighted by pre-period cell employment, with standard errors clustered by province-industry. Standard errors in parentheses.

Table S2: Client-geography-weighted dose: employment responses

	(1) BPO share	(2) log BPO	(3) log emp	(4) log emp, unwtd
Exposure × geography-weighted adoption (lag 1)	-0.1292	-0.0321		
	(0.0227)	(0.0112)		
Exposure × geography-weighted adoption (lag 1)			-0.0051	-0.0091
			(0.0022)	(0.0024)
Observations	1,479	1,312	67,290	67,832
R^2	0.977	0.953	0.960	0.902

Notes: Same specifications as main-text Table 1, columns 1 to 4, with the national adoption dose replaced by the client-geography-weighted dose at a one-quarter lag. The dose weights the US series at 70 percent, a UK block at 15 percent, and a Japan-Singapore block at 15 percent, following the buyer-geography mix of the Philippine IT-BPM market in the IBPAP Roadmap 2028. Columns and inference as in Table S1. Standard errors in parentheses.

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Table S3: The adoption dose series D_t, by quarter and construction segment

Quarter	D_t (% of US firms using AI)	Construction
2021Q1	0.00	assumed zero (pre-ChatGPT)
2021Q2	0.00	assumed zero (pre-ChatGPT)
2021Q3	0.00	assumed zero (pre-ChatGPT)
2021Q4	0.00	assumed zero (pre-ChatGPT)
2022Q1	0.00	assumed zero (pre-ChatGPT)
2022Q2	0.00	assumed zero (pre-ChatGPT)
2022Q3	0.00	assumed zero (pre-ChatGPT)
2022Q4	0.30	backcast: avg of Google Trends & Similarweb visit ratios to 2023Q1 x Ramp-based 2023Q1 value
2023Q1	1.78	backcast: BTOS 2023Q3 anchor x Ramp AI Index ratio
2023Q2	3.35	backcast: BTOS 2023Q3 anchor x Ramp AI Index ratio
2023Q3	3.70	BTOS original question (producing goods or services)
2023Q4	4.42	BTOS original question (producing goods or services)
2024Q1	4.74	BTOS original question (producing goods or services)
2024Q2	4.77	BTOS original question (producing goods or services)
2024Q3	5.47	BTOS original question (producing goods or services)
2024Q4	6.09	BTOS original question (producing goods or services)
2025Q1	7.33	BTOS original question (producing goods or services)
2025Q2	8.96	BTOS original question (producing goods or services)
2025Q3	9.38	BTOS original question (producing goods or services)
2025Q4	10.51	BTOS revised question x 0.600 wording-splice factor
2026Q1	11.09	BTOS revised question x 0.600 wording-splice factor
2026Q2	12.10	BTOS revised question x 0.600 wording-splice factor

Notes: D_t is the share of US firms using AI in production. The BTOS segment uses the US Census Bureau's biweekly survey aggregated to quarters. The backcast segment scales the first BTOS reading (3.7 percent, 2023Q3) by the growth of the Ramp AI adoption index. The rescaled segment places the revised BTOS question (in use from November 17, 2025) on the original basis using a factor of 0.600. The raw component series are included in the replication package.

S4 Tests of the Identifying Assumption

The estimates in the paper are identified by the curvature of the adoption path. The identifying assumption is that, conditional on the exposure-by-linear-trend interactions, no omitted shock moves employment across exposure groups with the same curvature. This section tests the two implications of that assumption that the pre-release data can assess. First, the pre-release exposure gradient should be linear in time, because any pre-existing curvature would imply the kind of omitted shock that the design rules out operating before the technology existed. I test this by adding exposure interacted with squared and cubed time to the seven pre-release rounds (2021Q2 to 2022Q4) and testing the added terms jointly. Second, a dose-shaped regressor placed where no dose existed should attract nothing. I assign the realized adoption curve of the first seven post-release quarters to the seven pre-release rounds and estimate each design with its linear trend control, so the placebo coefficient is identified from the same kind of curvature as the real one.

Table S4 reports both tests alongside each design's dose estimate. The two within-province designs pass both. In the industry design the curvature terms are jointly insignificant ($p = 0.79$)

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and the placebo dose attracts −0.002 (SE 0.005), a third of the real estimate and indistinguishable from zero. In the municipal design the corresponding figures are $p = 0.90$ and a wrong-signed 0.042 (SE 0.047). The province-level rows do not pass. The BPO employment share shows pre-release curvature ($p = 0.001$), and the placebo dose attracts −0.138 (SE 0.037), comparable in magnitude to the real estimate. The calendar explains why. The seven pre-release rounds span the reopening from pandemic restrictions, including renewed lockdowns in August 2021 and January 2022 that bound most tightly in the capital region, where exposure is highest, so the pre-release path of the exposed provinces is not linear. A curvature-based estimator at the province level therefore partly picks up the reopening cycle. The province-by-round fixed effects of the industry and municipal designs absorb the restriction cycle, and in those designs the pre-release period behaves as the identifying assumption requires. The province-level columns of main-text Table 1 should therefore be read as the descriptive gradient, while identification rests on the within-province and municipal designs.

Table S4: Tests of the identifying assumption, by design

	Dose estimate (lag 1)	Pre-release curvature (joint p-value)	Pre-release placebo dose
Province, BPO employment share	-0.1495 (0.0275)	0.001	-0.1380 (0.0369)
Province, log BPO employment	-0.0366 (0.0119)	0.088	-0.0738 (0.0433)
Industry within province, log employment	-0.0060 (0.0025)	0.792	-0.0022 (0.0051)
Municipality, log BPO employment	-0.0595 (0.0258)	0.898	0.0420 (0.0473)

Notes: The dose estimate is the coefficient on exposure interacted with US adoption at a one-quarter lag, as in main-text Table 1 (province columns weighted by pre-period employment, industry column weighted by pre-period cell employment, municipal column unweighted). The curvature column reports the joint p-value on exposure interacted with squared and cubed time, estimated on the seven pre-release rounds with the linear term included. The placebo column reports the coefficient on exposure interacted with the realized adoption curve of the first seven post-release quarters assigned to the seven pre-release rounds, with the linear trend control included, and its standard error in parentheses. Standard errors are clustered by province, province-industry, or municipality to match each design.

Panel (a) of main-text Figure 3 illustrates the dynamics in the design that passes these tests. It plots the within-province industry event-study coefficients net of a line fitted to the six estimated pre-release points. The detrended path sits at zero before the release, remains near zero in the first two quarters afterward, when lagged adoption was still close to zero, and turns negative in nearly every quarter from mid-2023 onward, matching the gradual accumulation implied by the dose model rather than the discrete break predicted by a release-date specification. The quarter-by-quarter deviations are individually imprecise because a trend extrapolated from six points carries substantial uncertainty, and the pooled post-release deviation of −0.004 (SE 0.018) reflects that extrapolation penalty. The powered test remains the joint dose-and-trend specification of main-text Table 1, which estimates the trend from all rounds jointly rather than extrapolating it.

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A shock specific to the exposed industries would survive the province-by-round fixed effects, and the principal candidate in this period is the return-to-office mandate for tax-zone firms. The Fiscal Incentives Review Board required registered firms to return workers onsite to retain their tax incentives, with the requirement binding from April 2023, a transfer window in late 2022 that allowed firms to retain remote work by re-registering with the Board of Investments, and a formal relaxation to half-remote work in November 2024. Three features distinguish this policy from the estimated adoption effects. First, its time path is a sequence of steps rather than the smooth accumulation that identifies the dose coefficient. Second, the November 2024 relaxation loosened the mandate while the employment response continued to strengthen through mid-2025. Third, the mandate provides no reason for firms to rotate employment away from the most exposed occupations within the sector, whereas Table S13 documents exactly that rotation. The mandate may have contributed to sectoral attrition in 2023, but it does not track the adoption path that identifies the estimates.

The deployment panel of Section S9 requires a different diagnostic, with the opposite conclusion about the trend control. In the province-by-quarter deployment panel, the partial correlation between the exposure-by-trend and exposure-by-dose interactions, after the fixed effects, is 0.82, and the correlation between the trend and the placebo dose is 0.94. The two regressors therefore split nearly the same variation: with the trend control included, shifting the dose series against the panel leaves the coefficient essentially unchanged, as negative at a two-year lead as at the design's own lag, so neither a trend-controlled estimate nor a placebo-dose test can date the shock. The deployment estimates therefore omit the trend control and instead place the identifying weight on the lead-lag profile in Panel (c) of the main-text adjustment figure. On a fixed estimation sample, the coefficient on log first-time deployments is null at leads of four to eight quarters, turns negative when the shock arrives, and grows monotonically thereafter. Total deployments fail this test, with large positive coefficients at long leads, indicating an exposure-correlated secular trend that first-time deployments do not exhibit. The analysis therefore focuses on first-time deployments and their share of total deployments.

Table S5 reports four additional checks on the deployment estimates. The first two address possible recording changes following the creation of the Department of Migrant Workers in 2022. Aggregating provinces to regions leaves the decline in place, and deployments from a province's neighboring regions fall rather than rise with its exposure, so no province's decline appears as a gain elsewhere. The remaining checks address the post-pandemic recovery, during which the most exposed provinces rebounded earliest. Excluding 2022, when this differential recovery is sharpest, leaves the estimate largely unchanged, while assigning the realized dose curve to 2018 and 2019, when the true dose was zero, yields a coefficient of similar magnitude but with a wide confidence interval. These checks indicate that the recorded decline in first-time deployments is not an artifact of changing administrative records. How much of that decline is attributable to the adoption dose remains less certain, and the main text reflects that qualification.

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Table S5: Deployment decline: recording and recovery checks

Check	Window	Coefficient	SE	[boot p]	Clusters
Region aggregation (17 regions)	2022q1–2025q2	-0.0237	(0.0056)	[0.088]	17
	2023q1–2025q2	-0.0146	(0.0014)	[0.007]	17
Region-mates of exposed province	2022q1–2025q2	-0.0185	(0.0026)	[0.003]	85
	2023q1–2025q2	-0.0113	(0.0010)	[0.000]	85
Rebound year excluded	2023q1–2025q2	-0.0126	(0.0021)	[0.000]	85
Placebo dose, pre-period	2018q1–2019q4	-0.0171	(0.0113)	[0.305]	85

Notes: Outcomes are log first-time land-based deployments. Region rows: the design of Table S10 on region aggregates, with exposure the employment-weighted regional average. Region-mates rows: the outcome is log deployments of the province's region excluding the province itself. Placebo row: the realized dose curve of 2022Q4–2024Q3 assigned to 2018Q1–2019Q4. Weighted by pre-period employment, wild-bootstrap p-values in brackets, 9,999 replications.

S5 Exposure-Robust Inference

Clustering by province leaves open the concern of Adão et al. (2019) that provinces with similar occupation mixes have correlated residuals. Following Borusyak et al. (2022), I re-estimate the long-difference form of the design at the occupation level, where the identifying assumption operates, using exposure-weighted averages of the outcome and the regressor and instrumenting with the occupation shocks. As the equivalence result implies, the occupation-level point estimate equals the province-level estimate. Table S6 shows that the exposure-robust standard errors are smaller than the province-clustered ones, so correlated-share dependence is not driving the paper's precision. The inverse Herfindahl of the occupation weights implies 14.5 effective shocks across 42 occupation groups. This concentration is inherent to the setting, because Philippine employment is concentrated in the clerical and customer-service occupations whose exposure the paper measures. One qualification applies when reading the table. The long-difference share outcome in columns 1 and 2 reflects the rising pre-period gradient in the share outcome, so the positive coefficient there is not an adoption effect. The table is included to assess inference rather than identification.

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Table S6: Shock-level inference, long-difference form

	(1) share, province	(2) share, shock-level	(3) log emp, province	(4) log emp, shock-level
Exposure (raw index)	14.18	14.19	-0.34	-0.34
	(2.37)	(1.16)	(0.81)	(0.23)
Units	87	42	87	42

Notes: The outcome is the change in the province mean from pre-release rounds (2021Q2 to 2022Q4) to post-release rounds (2023Q1 to 2025Q2), regressed on the raw exposure index and weighted by pre-period employment. Columns 1 and 3 report the province-level regression with heteroskedasticity-robust standard errors. Columns 2 and 4 report the numerically equivalent occupation-level regression with robust standard errors, which are valid under shock-level identification. Standard errors in parentheses.

S6 Municipal Design Details

The municipal analysis links the LFS workplace codes to the Philippine Standard Geographic Code, and the crosswalk matches 99.9 percent of the 1,645 workplace codes. Treatment intensity is the count of a municipality's IT park and IT center Presidential Proclamations through 2021. The proclamation panel is assembled from the full texts of 287 IT-zone proclamations, of which 97.9 percent are matched to a municipality and 90 percent carry an exact signing date, with the year of issue used for the remainder. The estimation sample retains municipalities observed with BPO employment in at least 12 of the 17 quarterly rounds, yielding a panel of roughly 1,600 municipality-round observations. Table S7 reports the estimates together with clustering variants for the headline province-level specification.

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Table S7: Municipal estimates and clustering variants

	(1) mun. log BPO	(2) mun. log BPO	(3) mun. unwtd	(4) prov. share	(5) prov. coarse cl.
IT zones (log) × trend	0.0228	0.0118	0.0367		
	(0.0162)	(0.0138)	(0.0148)		
IT zones (log) × AI adoption, lag 1	-0.0420		-0.0595		
	(0.0280)		(0.0258)		
IT zones (log) × kink		-0.0142			
		(0.0169)			
Exposure × trend				0.1202	0.1202
				(0.0118)	(0.0137)
Exposure × AI adoption, lag 1				-0.1495	-0.1495
				(0.0275)	(0.0333)
Observations	1,170	1,170	1,170	1,479	1,479
R^2	0.965	0.965	0.894	0.977	0.977

Notes: Columns 1 to 3 use the municipality-by-round panel of workplace BPO employment. IT-zone intensity is the log of one plus the count of IT park and IT center Presidential Proclamations through 2021. All municipal columns include municipality and province-by-round fixed effects, and standard errors are clustered by municipality. Columns 1 and 2 weight by pre-period municipal BPO employment. Columns 4 and 5 report the province-level dose specification with adoption at a one-quarter lag, clustered by province in column 4 and by ten island-group clusters in column 5. Standard errors in parentheses.

S7 Industry Exposure Ranking

The industry-level index assigns each two-digit PSIC division the exposure implied by its national pre-period occupation mix, using the same occupation weights as the provincial index. Table S8 reports the most and least exposed divisions. The ranking has face validity at both ends. Office administrative and business support services, which contains the call centers, is the most exposed division, followed by gambling and betting, which in the Philippines includes online-gaming operations staffed by customer-service and back-office workers, and then by information services, finance, head offices, and computer programming. The least exposed divisions are forestry, fishing, and other primary activities. The full ranking of all 86 divisions is included in the replication package.

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Table S8: Industry exposure, most and least exposed PSIC divisions

Rank	PSIC	Division	Exposure	Pre-period employment (millions)
1	82	Office administrative and business support (incl. call centres)	0.412	8.56
2	92	Gambling and betting activities	0.368	1.65
3	63	Information service activities	0.366	0.50
4	64	Financial services, except insurance	0.347	3.36
5	70	Head offices and management consultancy	0.328	0.29
6	62	Computer programming and consultancy	0.300	0.95
7	73	Advertising and market research	0.291	0.25
8	65	Insurance and pension funding	0.278	0.50
9	79	Travel agency, tour operator, and reservation services	0.265	0.26
10	66	Auxiliary financial activities	0.260	0.48
11	69	Legal and accounting activities	0.238	1.07
12	99	Extraterritorial organizations and bodies	0.225	0.01
13	78	Employment activities	0.221	0.42
14	58	Publishing activities	0.217	0.20
15	72	Scientific research and development	0.212	0.16
		⋮		
82	39	Remediation and other waste management services	0.011	0.02
83	38	Waste collection, treatment, and disposal	0.009	0.21
84	96	Other personal services	0.008	16.78
85	03	Fishing and aquaculture	0.006	9.15
86	02	Forestry and logging	0.005	0.84

Notes: Exposure is the employment-share-weighted mean of occupation-level observed AI exposure over each division's national pre-period (2021Q2 to 2022Q4) occupation mix. Employment is the division's average weighted employment over the pre-period rounds. Division titles follow PSIC 2009, which aligns with ISIC Revision 4.

S8 Predicted-Index Construction

The capability-based comparison index in Panel B of main-text Figure 1 replaces the observed-usage occupation weights with the AI Occupational Exposure scores of Felten et al. (2021) while holding every other convention fixed. The AIOE scores are published on 2010 SOC codes, so they enter the same crosswalk chain at the 2010 SOC stage, receive the same OEWS employment weights through the successor-code mapping, aggregate to the same two-digit ISCO groups, and combine with the same pre-period provincial employment shares. The resulting provincial index correlates with the observed-usage index at $r = 0.92$. The deviations are concentrated at the top of the distribution, where the observed-usage index ranks the Metro Manila districts higher than the capability-based index, suggesting that realized usage contains additional information where the BPO industry is concentrated.

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S9 Adjustment Margins

Table S9 reports the full grid of estimates underlying the adjustment-margins figure in the main text. The province block applies the main-text specification to log employment in each destination sector, with the BPO divisions 62, 63, and 82 excluded throughout, at adoption lags of one to four quarters, in both the full sample of 87 province-level areas and the sample excluding the four NCR districts. The full-sample agriculture coefficients illustrate why both samples are reported. The NCR districts carry large regression weights but contain agricultural cells of only a few hundred workers, so their log employment is dominated by sampling noise. Excluding them changes the lag-1 agriculture coefficient from 0.117 to 0.011. No other coefficient changes sign or becomes statistically distinguishable from zero across the two samples.

The region block estimates the same specification across the 17 regions, replacing provincial exposure with its employment-weighted regional average. Unemployment, labor force participation, and overseas membership are attributes of residence rather than workplace, and the public-use files identify residence only at the regional level, making the region the finest geography at which these margins can be measured. Employment status comes from the LFS summary-status variable and overseas membership from the household roster indicator for overseas Filipino workers, aggregated by region and round using survey weights. The resulting series are included in the replication package, and the national unemployment and participation series reported in the main text are obtained by aggregating these regional series.

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Table S9: Adjustment margins: full grid of dose estimates

Outcome	Sample	Lag 1	Lag 2	Lag 3	Lag 4	N
Province level: log sector employment						
Agriculture	all areas	0.1170 (0.0237)	0.1069 (0.0211)	0.0336 (0.0085)	0.0367 (0.0436)	1,476
	excl. NCR	0.0111 (0.0153)	0.0317 (0.0247)	0.0318 (0.0261)	0.0292 (0.0260)	1,411
Manufacturing	all areas	-0.0028 (0.0098)	0.0002 (0.0092)	0.0014 (0.0070)	-0.0129 (0.0054)	1,479
	excl. NCR	-0.0314 (0.0274)	-0.0337 (0.0279)	-0.0150 (0.0218)	-0.0212 (0.0183)	1,411
Other industry	all areas	-0.0156 (0.0062)	-0.0136 (0.0053)	-0.0116 (0.0046)	-0.0188 (0.0044)	1,479
	excl. NCR	-0.0314 (0.0203)	-0.0263 (0.0232)	-0.0143 (0.0204)	-0.0208 (0.0178)	1,411
Trade	all areas	0.0078 (0.0032)	0.0086 (0.0034)	0.0051 (0.0035)	-0.0048 (0.0051)	1,479
	excl. NCR	0.0004 (0.0132)	-0.0021 (0.0108)	-0.0079 (0.0116)	-0.0252 (0.0116)	1,411
Transport	all areas	0.0032 (0.0049)	0.0004 (0.0046)	-0.0026 (0.0040)	-0.0072 (0.0047)	1,479
	excl. NCR	-0.0057 (0.0181)	-0.0047 (0.0143)	0.0084 (0.0158)	0.0180 (0.0138)	1,411
Accommodation and food	all areas	-0.0054 (0.0074)	0.0005 (0.0093)	0.0087 (0.0075)	0.0130 (0.0101)	1,462
	excl. NCR	-0.0211 (0.0301)	-0.0188 (0.0367)	0.0027 (0.0311)	0.0296 (0.0398)	1,394
Market services (excl. BPO)	all areas	-0.0166 (0.0086)	-0.0213 (0.0062)	-0.0176 (0.0063)	-0.0142 (0.0077)	1,475
	excl. NCR	0.0080 (0.0255)	-0.0211 (0.0280)	-0.0299 (0.0229)	-0.0250 (0.0268)	1,407
Public and social services	all areas	0.0028 (0.0037)	0.0017 (0.0028)	-0.0002 (0.0049)	-0.0028 (0.0036)	1,479
	excl. NCR	0.0204 (0.0140)	0.0052 (0.0137)	0.0021 (0.0156)	0.0031 (0.0176)	1,411
Other services	all areas	-0.0150 (0.0049)	-0.0103 (0.0058)	0.0054 (0.0072)	0.0045 (0.0054)	1,477
	excl. NCR	-0.0081 (0.0178)	0.0151 (0.0209)	0.0558 (0.0167)	0.0452 (0.0146)	1,409
Total non-BPO	all areas	0.0004 (0.0019)	0.0030 (0.0025)	-0.0014 (0.0025)	-0.0061 (0.0022)	1,479
	excl. NCR	0.0044 (0.0059)	0.0143 (0.0070)	0.0070 (0.0066)	0.0009 (0.0058)	1,411
Total employment	all areas	-0.0013 (0.0020)	0.0015 (0.0025)	-0.0024 (0.0021)	-0.0068 (0.0019)	1,479
	excl. NCR	0.0041 (0.0060)	0.0138 (0.0071)	0.0058 (0.0071)	-0.0004 (0.0062)	1,411
Region level: labor market margins						
Unemployment rate	all regions	0.2852 (0.0409)	0.0807 (0.0520)	0.1202 (0.0563)	0.1098 (0.0553)	282
Labor force participation	all regions	0.2713 (0.2201)	0.3535 (0.3310)	0.1563 (0.2179)	0.0011 (0.1264)	282
OFWs per 1,000 residents	all regions	-0.0981 (0.0860)	0.0551 (0.1206)	0.0954 (0.1156)	0.1290 (0.1525)	282

Notes: Each cell reports the coefficient on exposure interacted with US adoption at the indicated lag, from the main-text specification with exposure-by-trend interactions and unit and round fixed effects, with the standard error in parentheses. Province rows: log employment in the indicated sector group, weighted by pre-period employment, standard errors clustered by province, BPO divisions excluded. Region rows: unemployment rate and labor force participation in percentage points and overseas Filipino workers per 1,000 residents, exposure replaced by its employment-weighted regional average, weighted by pre-period employment, standard errors clustered by region. N refers to the lag-1 estimate.

The overseas flow estimates use administrative records from the Department of Migrant Workers covering monthly counts of deployed land-based workers by province of origin from January 2018 through June 2026, separately identifying first-time hires and rehires, with annual breakdowns by age and sex from 2022 and by destination from 2018. The 85 provinces are matched to the exposure index by name (the build scripts document the small number of label corrections), and the breakdown sheets reconcile exactly to the monthly totals. Two features matter. First, deployment is recorded at the province of origin whereas exposure is measured at the province of workplace,

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so this mismatch attenuates the estimates. Second, a revision to new-hire recording from June 2025 lowers counts by roughly one-third in every province. The quarterly analysis therefore ends in 2025Q2, and the annual designs absorb the break through year or province-by-year fixed effects. The workbook itself is not publicly available and is therefore not included in the replication package, though researchers may request it from the same office. The parsed province-level panels are included.

Table S10 reports the province-by-quarter estimates, with province-by-quarter-of-year and year-quarter fixed effects, the adoption dose lagged two quarters, pre-period employment weights, and province-clustered standard errors. The trend-control columns are included for completeness. As discussed in Section S4, the trend and dose are nearly collinear in this panel, so those specifications cannot date the shock, and identification instead rests on the lead-lag profile shown in Panel (c) of the main-text adjustment figure.

Table S10: Deployment flows by province of origin: dose estimates

		No trend control			With trend control		
Outcome	Window	Coef.	SE	[boot p]	Coef.	SE	[boot p]
New hires	2021q2–2025q2	-0.0171	(0.0027)	[0.000]	-0.0303	(0.0079)	[0.028]
	2022q1–2025q2	-0.0201	(0.0036)	[0.004]	-0.0098	(0.0030)	[0.011]
Rehires	2021q2–2025q2	0.0048	(0.0019)	[0.207]	-0.0125	(0.0058)	[0.054]
	2022q1–2025q2	0.0013	(0.0022)	[0.648]	0.0033	(0.0033)	[0.211]
New-hire share	2021q2–2025q2	-0.0043	(0.0006)	[0.001]	-0.0002	(0.0013)	[0.846]
	2022q1–2025q2	-0.0035	(0.0006)	[0.002]	-0.0030	(0.0010)	[0.020]
Total deployment	2021q2–2025q2	0.0044	(0.0013)	[0.009]	-0.0314	(0.0094)	[0.033]
	2022q1–2025q2	-0.0012	(0.0017)	[0.435]	-0.0079	(0.0033)	[0.029]

Notes: Outcomes are log deployed land-based workers by province of origin, in a province-by-quarter panel of 85 provinces. Each cell is the coefficient on exposure interacted with the dose lagged two quarters, with province-by-quarter-of-year and year-quarter fixed effects, weighted by pre-period employment. The new-hire share is first-time hires over all deployments. Standard errors clustered by province, wild-bootstrap p-values in brackets, 9,999 replications. Monthly and unweighted variants yield the same pattern.

Tables S11 and S12 report the annual breakdowns underlying the mechanism discussed in the main text. Panel A of each table estimates the levels specification on the annual panel (2022–2025), with province and year fixed effects and the dose averaged over the preceding year. Panel B implements a triple-difference specification across groups within province-years. Province-by-year fixed effects absorb every province-level confound, including the trend collinearity and the 2025 recording break, so the coefficients ask only whether the decline is concentrated in particular groups. It is not. The decline is as large among older workers as younger ones, no larger for women than men, and statistically indistinguishable across destination corridors, including those whose employment shares little with BPO work. This uniformity motivates the household-income interpretation in the main text.

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Table S11: New-hire deployments by age bracket and sex: dose estimates

	Coefficient	SE	[boot p]
Panel A: dose effect by age bracket (province and year fixed effects)			
25 and below	-0.0241	(0.0058)	[0.001]
26–30	-0.0132	(0.0028)	[0.000]
31–35	-0.0158	(0.0036)	[0.004]
36–40	-0.0163	(0.0030)	[0.013]
41–45	-0.0182	(0.0033)	[0.002]
46–50	-0.0250	(0.0046)	[0.013]
51 and up	-0.0247	(0.0063)	[0.018]
Panel B: differential by group (province-by-year fixed effects, base 41 and up)			
30 and below × dose	0.0031	(0.0022)	[0.254]
31–40 × dose	0.0033	(0.0010)	[0.067]
Female × dose (base: male)	0.0019	(0.0012)	[0.121]

Notes: Outcomes are log first-time land-based deployments by province and age bracket (Panel A) or group (Panel B), annual, 2022–2025. Panel A: exposure interacted with the lagged annual dose, province and year fixed effects. Panel B: group-specific dose interactions under province-by-year, group-by-year, and group-by-province fixed effects, identified within province-year, with ages 41 and above and men the omitted categories. Weighted by pre-period employment, clustered by province, wild-bootstrap p-values in brackets. Excluding 2025 leaves Panel B unchanged.

Table S12: New-hire deployments by destination: dose estimates

	Coefficient	SE	[boot p]
Panel A: dose effect by destination corridor (province and year fixed effects)			
Hong Kong, Macau, Singapore (domestic work)	-0.0178	(0.0034)	[0.013]
Gulf and Middle East (domestic work, construction)	-0.0147	(0.0032)	[0.003]
Taiwan, Japan, Korea (production, care)	-0.0154	(0.0031)	[0.002]
Western economies (professional, nursing)	-0.0174	(0.0041)	[0.033]
All other destinations	-0.0131	(0.0043)	[0.003]
Panel B: differential by corridor (province-by-year fixed effects, base Hong Kong–Macau–Singapore)			
Gulf × dose	0.0028	(0.0028)	[0.490]
Taiwan–Japan–Korea × dose	0.0024	(0.0035)	[0.518]
Western × dose	0.0001	(0.0048)	[0.983]
Other × dose	0.0043	(0.0041)	[0.489]

Notes: Outcomes are log first-time land-based deployments by province and destination corridor, annual, 2022–2025. Corridors group destinations by the dominant occupation mix of newly hired workers. Panel A: exposure interacted with the lagged annual dose, province and year fixed effects. Panel B: corridor-specific dose interactions under province-by-year, corridor-by-year, and corridor-by-province fixed effects. Weighted by pre-period employment, clustered by province, wild-bootstrap p-values in brackets. Estimates for the eight largest single destinations range from -0.006 (Kuwait) to -0.032 (Saudi Arabia), seven of eight significant, with no ordering by skill content.

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A related question is whether adjustment occurs within the BPO sector itself, shifting employment from more exposed to less exposed work. Table S13 reports two tests. The division rows restrict the industry panel to the three BPO divisions and interact division exposure with the dose under province-by-round and province-by-division fixed effects. The occupation rows construct a national occupation-by-round panel within the BPO divisions and interact occupation exposure with the dose under occupation and round fixed effects. The occupation mix rotates away from high-exposure occupations under both weighting schemes, indicating that part of the adjustment occurs within the sector and that the main-text employment estimates net out this reallocation. The division estimates differ across weighting schemes because employment outside the major hubs is sparse in the programming and information divisions.

Table S13: Within-BPO reallocation: dose estimates

	Coefficient	SE	*N*
Divisions within provinces, weighted	-0.0487	0.0234	2,293
Divisions within provinces, unweighted	0.0822	0.0297	2,312
Occupations within the sector, weighted	-0.0733	0.0250	403
Occupations within the sector, unweighted	-0.0407	0.0207	405

Notes: Each cell reports the coefficient on exposure interacted with US adoption at a one-quarter lag, with exposure-by-trend interactions included. Division rows: log employment in BPO divisions 62, 63, and 82 by province and round, with province-by-round and province-by-division fixed effects, standard errors clustered by province-division, weighted by pre-period cell employment where indicated. Occupation rows: log national employment by two-digit occupation within the BPO divisions, occupation and round fixed effects, standard errors clustered by occupation, weighted by pre-period occupation employment where indicated.

A policy discussion follows from the adjustment mechanism rather than the point estimates themselves. If the displacement documented here continues as adoption spreads, three broad responses are available. The first is to redirect the industry toward less exposed segments, such as healthcare information management, complex back-office work, and software services. Table S13 suggests that this adjustment has already begun, though the less exposed segments remain substantially smaller than the work they would need to replace. The second is to expand domestic employment opportunities outside the sector, although the relative weakness of these alternatives is precisely what made exported services the country's development strategy. The third is overseas employment.

The overseas margin is the most immediately available because its institutions already exist. The Department of Migrant Workers negotiates bilateral labor agreements, licenses recruitment agencies, and administers deployment procedures, so expanding overseas opportunities is a policy lever that can move more quickly than domestic industrial transformation. Agreements matched to the skills of displaced service workers, together with policies that reduce the upfront cost of migration, would widen this adjustment margin if household financing constraints are limiting departures.

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The overseas option nonetheless has limits. Destination countries are adopting the same technologies, so future labor demand is likely to concentrate in work requiring physical presence rather than the office occupations most exposed to generative AI. Overseas migration can therefore provide a bridge for some displaced workers, but it is unlikely to substitute fully for the labor demand that exported services once generated.

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